APM1110_FA8_Valdez_ChanceJayden.pdf

https://github.com/ChanceValdez/APM1110-FA8

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README

These are the typeset solutions to Formative Assessment 8 of APM1110 (Probability and Probability Distributions) at Far Eastern University.

The solutions involve R, which I used to speed up the repetitive computation of combinations, along with calculus, proof-writing concepts, and, above all, probability concepts, as the course name implies.

My handwritten scratch work is also in the GitHub repository as proof that I did these solutions independently. I typeset them as a second pass to double-check my work. The repository contains everything: the LaTeX code, the R code, the handwritten scratch work, and the compiled PDF.

Note: The final solutions are the ones in this typeset document. The handwritten solutions may lack insight and some of what I want to convey, and my handwriting is of questionable quality. Please grade the compiled PDF, since it holds my final solutions, and use the handwritten work only to verify that I did them independently.

Assessment Items (30 Points)

1. (6 points) An urn contains 10 red, 8 black, and 7 green balls. One of the colors is chosen at random (meaning that the chosen color is equally likely to be any of the 3 colors), and then 4 balls are randomly chosen from the urn. Let X be the number of these balls that are of the chosen color.

(a) (2 points) Find $P(X = 0)$.

Let C be the chosen color. By the law of total probability,

$$P(X=0) = \sum_{c \in \{\text{Red, Black, Green}\}} P(X=0 \mid C=c)\, P(C=c).$$

Given $C = c$, $X = 0$ means that all 4 balls are not of the chosen color, c. Since there are $10 + 8 + 7 = 25$ balls in total, the total number of equally likely draws is

$$\binom{25}{4} = 12650.$$

Counting the draws that avoid each color (computed in R with `choose()`):

$$P(X=0 \mid \text{Red}) = \frac{\binom{15}{4}}{12650} = \frac{1365}{12650}, \quad P(X=0 \mid \text{Black}) = \frac{\binom{17}{4}}{12650} = \frac{2380}{12650}, \quad P(X=0 \mid \text{Green}) = \frac{\binom{18}{4}}{12650} = \frac{3060}{12650}.$$

R Code used for the Combinations:

```
total_ways <- choose(25, 4)
ways_if_red <- choose(15, 4)
ways_if_black <- choose(17, 4)
ways_if_green <- choose(18, 4)

print(total_ways)
## [1] 12650
print(ways_if_red)
```

```
## [1] 1365
print(ways_if_black)
## [1] 2380
print(ways_if_green)
## [1] 3060
```

Weighting each color by $P(C = c) = \frac{1}{3}$:

$$P(X = 0) = \frac{1}{3}\left[\frac{1365 + 2380 + 3060}{12650}\right] = \frac{1361}{7590} \approx 0.1793.$$

$\boxed{\text{Therefore: } P(X = 0) \approx 0.1793}$

(b) (2 points) Let X_i equal 1 if the ith ball selected is of the chosen color, and let it equal 0 otherwise. Find:

$$P(X_i = 1), \quad i = 1, 2, 3, 4.$$

By symmetry, the ith ball is a uniformly random ball from the urn. So, given a color, $P(X_i = 1 \mid C = c)$ is that color's fraction of the 25 balls. Weighting each color by $\frac{1}{3}$:

$$P(X_i = 1) = \frac{1}{3}\left[\frac{10}{25} + \frac{8}{25} + \frac{7}{25}\right] = \frac{1}{3}.$$

This holds because no matter what color the ith ball turns out to be, our independent guess C has a $\frac{1}{3}$ chance of matching it. Thus,

$$\boxed{P(X_i = 1) = \frac{1}{3} \quad \text{for } i = 1, 2, 3, 4.}$$

(c) (2 points) Find $E[X]$.

Hint: Express X in terms of X_1, X_2, X_3, and X_4.

Linearity of Expectation states that for any random variables $Y_1, \ldots, Y_n$ (independent or not),

$$E[Y_1 + Y_2 + \cdots + Y_n] = E[Y_1] + E[Y_2] + \cdots + E[Y_n].$$

Since $X = X_1 + X_2 + X_3 + X_4$, linearity gives

$$E[X] = E[X_1] + E[X_2] + E[X_3] + E[X_4].$$

Each X_i is an indicator, so $E[X_i] = P(X_i = 1) = \frac{1}{3}$ by (b). Linearity applies even though the draws are not independent. Hence

$$E[X] = E[X_1] + E[X_2] + E[X_3] + E[X_4] = \frac{1}{3} + \frac{1}{3} + \frac{1}{3} + \frac{1}{3} = \boxed{\frac{4}{3} \approx 1.3333}.$$

2. (8 points) The random variable X is said to have the Yule-Simon distribution if

$$P(X = n) = \frac{4}{n(n+1)(n+2)}, \quad n \geq 1.$$

(a) (3 points) Show that the preceding is actually a probability mass function. That is, show that

$$\sum_{n=1}^{\infty} P(X = n) = 1.$$

Hint: For (a), first use that

$$\frac{1}{n(n+1)(n+2)} = \frac{1}{2}\left[\frac{1}{n(n+1)} - \frac{1}{(n+1)(n+2)}\right],$$

then use that

$$\frac{k}{n(n+k)} = \frac{1}{n} - \frac{1}{n+k}.$$

Proof. A Probability Mass Function (PMF) requires the following:

i. Every value must be nonnegative.

ii. The values must add up to 1.

Nonnegativity: For $n \geq 1$,

$$P(X = n) = \frac{4}{n(n+1)(n+2)} > 0.$$

Total Sum: Substituting and using the hint,

$$\sum_{n=1}^{\infty} P(X = n) = 4\sum_{n=1}^{\infty} \frac{1}{n(n+1)(n+2)} = 4\sum_{n=1}^{\infty} \frac{1}{2}\left[\frac{1}{n(n+1)} - \frac{1}{(n+1)(n+2)}\right] = 2\sum_{n=1}^{\infty}\left[\frac{1}{n(n+1)} - \frac{1}{(n+1)(n+2)}\right].$$

Let $S_N = \sum_{n=1}^{N} P(X = n)$. Then

$$S_N = 2\left[\left(\frac{1}{2} - \frac{1}{6}\right) + \left(\frac{1}{6} - \frac{1}{12}\right) + \left(\frac{1}{12} - \frac{1}{20}\right) + \cdots + \left(\frac{1}{N(N+1)} - \frac{1}{(N+1)(N+2)}\right)\right].$$

Each term's second part cancels the next term's first part (telescoping), so only the first and last pieces remain:

$$S_N = 2\left[\frac{1}{2} - \frac{1}{(N+1)(N+2)}\right].$$

An infinite series is the limit of its partial sums as $N \to \infty$, and since $\frac{1}{(N+1)(N+2)} \to 0$,

$$\sum_{n=1}^{\infty} P(X = n) = \lim_{N\to\infty} S_N = 2 \cdot \frac{1}{2} = 1.$$

Since $P(X = n) > 0$ for all $n \geq 1$ and the total sum equals 1, the Yule-Simon distribution is a PMF. ■

(b) (2 points) Show that

$$E[X] = 2.$$

Proof. To begin, we use the definition of expected value for discrete random variables, $E[X] = \sum_{n=1}^{\infty} n\, P(X = n)$. Substituting and canceling n,

$$\sum_{n=1}^{\infty} n\, P(X = n) = \sum_{n=1}^{\infty} \frac{4n}{n(n+1)(n+2)} = 4\sum_{n=1}^{\infty} \frac{1}{(n+1)(n+2)}.$$

Using Partial Fraction Decomposition, write $\frac{1}{(n+1)(n+2)} = \frac{A}{n+1} + \frac{B}{n+2}$. Then $1 = A(n+2) + B(n+1)$, which gives $A = 1$ and $B = -1$, so

$$4\sum_{n=1}^{\infty}\left[\frac{1}{n+1} - \frac{1}{n+2}\right].$$

Let $S_N = \sum_{n=1}^{N} n\, P(X = n)$. Then

$$S_N = 4\left[\left(\frac{1}{2} - \frac{1}{3}\right) + \left(\frac{1}{3} - \frac{1}{4}\right) + \cdots + \left(\frac{1}{N+1} - \frac{1}{N+2}\right)\right] = 4\left[\frac{1}{2} - \frac{1}{N+2}\right].$$

Since $\frac{1}{N+2} \to 0$ as $N \to \infty$,

$$E[X] = \lim_{N\to\infty} S_N = 4 \cdot \frac{1}{2} = 2. \quad ■$$

(c) (3 points) Show that

$$E[X^2] = \infty.$$

Proof. For a discrete random variable, $E[X^2] = \sum_{n=1}^{\infty} n^2\, P(X = n)$. Substituting and canceling one factor of n,

$$E[X^2] = \sum_{n=1}^{\infty} \frac{4n^2}{n(n+1)(n+2)} = \sum_{n=1}^{\infty} \frac{4n}{(n+1)(n+2)} = 4\sum_{n=1}^{\infty} \frac{n}{n^2+3n+2}.$$

We use the Limit Comparison Test, which states that if $\sum a_n$ and $\sum b_n$ are series with positive terms and $\lim_{n\to\infty} \frac{a_n}{b_n} = c$, where c is finite and $c > 0$, then either both series converge or both diverge.

Let $a_n = \dfrac{n}{n^2+3n+2}$ and $b_n = \dfrac{1}{n}$ (the harmonic series, which diverges). Both have positive terms for $n \geq 1$. Then

$$\lim_{n\to\infty} \frac{a_n}{b_n} = \lim_{n\to\infty} \frac{n^2}{n^2+3n+2}.$$

This is of the form $\frac{\infty}{\infty}$, so applying L'Hôpital's rule twice,

$$\lim_{n\to\infty} \frac{2n}{2n+3} = \lim_{n\to\infty} \frac{2}{2} = 1.$$

Since $c = 1$ is finite and greater than 0, and $\sum \frac{1}{n}$ diverges, $\sum a_n$ also diverges. Multiplying a divergent series by 4 keeps it divergent. Therefore, $E[X^2] = 4\sum a_n = \infty$. ■

3. (4 points) The probability density function (PDF) of X is given by

$$f(x) = \begin{cases} a + bx^2, & 0 \leq x \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

If $E[X] = \frac{3}{5}$, find the values of a and b.

Solution. We are given the following:

1. f is a PDF on $[0,1]$ and 0 otherwise.
2. $E[X] = \frac{3}{5}$

Equation 1. For f to be a PDF, its integral over $[0,1]$ must equal 1:

$$\int_0^1 (a+bx^2)\,dx = \left[ax + \frac{b}{3}x^3\right]_0^1 = a + \frac{b}{3} = 1.$$

Equation 2. By the definition of $E[X]$ for a continuous random variable,

$$E[X] = \int_0^1 x(a+bx^2)\,dx = \left[\frac{a}{2}x^2 + \frac{b}{4}x^4\right]_0^1 = \frac{a}{2} + \frac{b}{4} = \frac{3}{5}.$$

From Equation 1, $a = 1 - \frac{b}{3}$. Substituting this into Equation 2 gives

$$\begin{aligned} \frac{1-\frac{b}{3}}{2} + \frac{b}{4} &= \frac{3}{5} \\ \frac{3-b}{6} + \frac{b}{4} &= \frac{3}{5} \\ 10(3-b) + 15b &= 36 \\ 30 - 10b + 15b &= 36 \\ 30 + 5b &= 36 \\ 5b = 6 \implies \boxed{b = \frac{6}{5}}. \end{aligned}$$

Substituting $b = \frac{6}{5}$ back into $a = 1 - \frac{b}{3}$:

$$a = 1 - \frac{\frac{6}{5}}{3} = 1 - \frac{6}{15} = 1 - \frac{2}{5} = \boxed{\frac{3}{5}}.$$

Check. Equation 1: $\frac{3}{5} + \frac{1}{3}\cdot\frac{6}{5} = \frac{3}{5} + \frac{2}{5} = 1$. Equation 2: $\frac{1}{2}\cdot\frac{3}{5} + \frac{1}{4}\cdot\frac{6}{5} = \frac{3}{10} + \frac{3}{10} = \frac{3}{5}$. Also, $f(x) \geq 0$ on $[0,1]$ since $a, b > 0$ and $x^2 \geq 0$, so f is a valid PDF.

Therefore, $\boxed{a = \frac{3}{5}, \quad b = \frac{6}{5}}$.

4. (4 points) Let X be a random variable having expected value μ and variance σ^2. Find the expected value and variance of

$$Y = \frac{X - \mu}{\sigma}.$$

Solution. By definition, $\mu = E[X]$ and $\sigma^2 = \mathrm{Var}(X) = E[(X-\mu)^2]$, with $\sigma > 0$. Rewrite Y in the form $aX + b$:

$$Y = \frac{1}{\sigma}X - \frac{\mu}{\sigma}, \qquad a = \frac{1}{\sigma},\ b = -\frac{\mu}{\sigma}.$$

Using the linear transformation rule for expectation, $E[aX + b] = aE[X] + b$:

$$E[Y] = \frac{1}{\sigma}E[X] - \frac{\mu}{\sigma} = \frac{\mu}{\sigma} - \frac{\mu}{\sigma} = 0.$$

Using the linear transformation rule for variance, $\mathrm{Var}(aX + b) = a^2\,\mathrm{Var}(X)$:

$$\mathrm{Var}(Y) = \left(\frac{1}{\sigma}\right)^2 \sigma^2 = 1.$$

Therefore, $\boxed{E[Y] = 0 \text{ and } \mathrm{Var}(Y) = 1}$.

5. (8 points) Let X and Y be independent Poisson random variables with parameters $\lambda_1 = 2$ and $\lambda_2 = 3$, respectively. Let $W = X + Y$.

(a) (3 points) Using the property of sums of independent random variables, derive the moment generating function (MGF) $M_W(t)$ of W.

First, let Z be a Poisson random variable with parameter λ. Its moment generating function is derived in Example 7b of my instructor's lecture slides (*M5. Functions of Random Variables*, Slide 49), and is reproduced here:

$$M_Z(t) = E[e^{tZ}] = \sum_{k=0}^{\infty} e^{tk}\,\frac{e^{-\lambda}\lambda^k}{k!} = e^{-\lambda}\sum_{k=0}^{\infty}\frac{(\lambda e^t)^k}{k!} = e^{-\lambda}e^{\lambda e^t} = e^{\lambda(e^t-1)}.$$

Substituting $\lambda_1 = 2$ and $\lambda_2 = 3$ gives the individual MGFs of X and Y respectively:

$$M_X(t) = e^{2(e^t-1)}, \qquad M_Y(t) = e^{3(e^t-1)}.$$

Since X and Y are independent, the moment generating function of the sum is the product of the individual moment generating functions:

$$M_W(t) = M_X(t)\cdot M_Y(t) = e^{2(e^t-1)}\cdot e^{3(e^t-1)} = \boxed{e^{5(e^t-1)}}$$

(b) (2 points) Based on the uniqueness property of MGFs, identify the probability distribution and parameter(s) of W.

The uniqueness property states that if a moment generating function exists and is finite in some region about $t = 0$, then it uniquely determines the distribution. Comparing $M_Z(t) = e^{\lambda(e^t-1)}$ with $M_W(t) = e^{5(e^t-1)}$, they match with $\lambda = 5$. Therefore, $\boxed{W \text{ has a Poisson(5) distribution}}$.

(c) (3 points) Determine the expected value $E[W]$ and the second moment $E[W^2]$ by successively differentiating $M_W(t)$ and evaluating it at $t = 0$.

Differentiating $M_W(t) = e^{5(e^t-1)}$ with the Chain Rule ($\frac{d}{dt}e^u = e^u \cdot u'$):

$$\frac{d}{dt}\,e^{5(e^t-1)} = e^{5(e^t-1)}\cdot\frac{d}{dt}\left[5(e^t-1)\right] = 5e^t\,e^{5(e^t-1)}.$$

Evaluating this at $t = 0$ gives the expected value:

$$E[W] = \left.\frac{d}{dt}M_W(t)\right|_{t=0} = 5(1)\,e^0 = 5.$$

Differentiating again with the Product Rule ($\frac{d}{dt}(uv) = u'v + uv'$, with $u = 5e^t$ and $v = e^{5e^t-5}$):

$$\frac{d}{dt}\left[\left(5e^t\right)\left(e^{5e^t-5}\right)\right] = \left(5e^t\right)\left(5e^t\right)\left(e^{5e^t-5}\right) + \left(5e^t\right)\left(e^{5e^t-5}\right) = 25e^{2t}\left(e^{5e^t-5}\right) + \left(5e^t\right)\left(e^{5e^t-5}\right).$$

Evaluating this at $t = 0$ gives the second moment:

$$E[W^2] = \frac{d^2}{dt^2}M_W(t)\bigg|_{t=0} = 25(1)\left(e^{5(1)-5}\right) + 5(1)\left(e^{5(1)-5}\right) = 25 + 5 = 30.$$

Check. A Poisson random variable has a mean equal to its variance: $\mathrm{Var}(W) = E[W^2] - (E[W])^2 = 30 - 5^2 = 5 = E[W]$.

Therefore, $\boxed{E[W] = 5}$ and $\boxed{E[W^2] = 30}$.